ARUNJYOTI SARMA

Doctoral Research Scholar | Indian Institute of Technology (BHU), Varanasi

FPGA Systems Architect • Flexible Electronics & Micro-Sensors • Embedded Systems & Edge AI

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GitHub: github.com/arunjyotisarmarsece24-jpg/neuroquant-lob | **Nationality:** Indian

1. EXECUTIVE SUMMARY & RESEARCH PROFILE

Hardware Systems Engineer and Doctoral Scholar at the Department of Electronics Engineering, **Indian Institute of Technology (BHU) Varanasi**, specializing in flexible electronics, strain-sensing structural health monitoring (SHM), and ultra-low-latency FPGA digital architectures. Demonstrated track record of translating advanced multi-physics modeling and analog sensor conditioning into high-performance silicon: designed and physically closed a 322 MHz Spartan-7 FPGA deterministic limit-order-book engine (sub-microsecond latency, zero DSP, 82 mW power), pioneered flexible PVDF piezoelectric biomechanical energy harvesting (lead author, IEEE-ECBIOS 2026), and architected novel computing IP (USR-X gear-shift BNN/MAC register; VEGA RISC-V Processing-in-Memory cancer diagnostic accelerator for DVCon India 2025). Combines 8+ years of dedicated self-directed engineering R&D;, continuous national engineering excellence (four-time GATE qualifier in EC and EE: 2019, 2021, 2023, 2024), and premier doctoral research at India's foremost technical institutions.

2. EDUCATION & ACADEMIC APPOINTMENTS

Ph.D. in Electronics Engineering (Doctoral Scholar) — *IIT (BHU), Varanasi* *July 2024 – Present*

- **Doctoral Research:** Flexible electronics, strain-sensing arrays for structural health monitoring (SHM), piezoelectric nanogenerators (PENG), analog front-end readout circuits, and embedded digital telemetry.
- **Completed Coursework (54 Total Credits):** Signal Processing (EC-511), Information Theory & Coding (EC-512), Neural Networks (EC-523), Microprocessor Engineering & Applications (EC-521), Mobile Communication & Wireless Networks (EC-513), Switching Theory & Logic Design (EC-522).
- **Doctoral Fellowship:** Conferred full-time Institute Research Fellowship by the Ministry of Education (MoE), Government of India.

B.Tech. in Electronics & Electrical Engineering — *KIIT University, Bhubaneswar* *2012 – 2016*

- **Academic Standing:** CGPA: **7.51 / 10.0** (First Class with Distinction). Active Member, KIIT Robotics Society.
- **Core Domains:** Power Electronics, Microprocessors & Microcontrollers, Analog & Digital VLSI, Control Systems, Signals & Systems, C/C++.
- **Undergraduate Capstone:** Industrial Induction Motor Control via DTMF Telemetry; Autonomous Renewable Energy Generation Systems.

3. PROFESSIONAL & ADVANCED ENGINEERING RESEARCH EXPERIENCE

Independent Systems & Hardware R&D; Engineer — *Applied Research & Prototyping Laboratory* *June 2016 – June 2024*

- Executed an 8-year sustained research program in digital RTL acceleration, embedded architectures, and rigorous competitive engineering benchmarks; qualified national GATE examinations in four distinct cycles (2019, 2021, 2023, 2024) across EC and EE branches.
- **IP Architecture (USR-X):** Invented and validated a multi-mode gear-shift Universal Shift Register featuring barrel network stride shifting, LFSR PRPG/MISR logic built-in self-test (LBIST), and dual-mode compute agility (bit-serial MAC for MLPs and XNOR+popcount BNNs).
- **Embedded Systems & Firmware:** Developed custom firmware for ARM Cortex, ATmega, and PIC microcontrollers for industrial motor drives, acoustic sensing, and wireless telemetry.
- **Premier Institute Selection:** Shortlisted and defended technical research proposals across top IITs (IIT Roorkee, IIT Delhi, IIT Guwahati, IISc Bangalore, IIT Kanpur, IIT Madras), culminating in selection for the Ph.D. program at IIT (BHU) Varanasi.

Academic & Technical Mentor (Electrical & Electronics Engineering) — *Engineering Foundations* *August 2017 – December 2023*

- Mentored aspiring engineering scholars in Microprocessors, Power Electronics, Circuit Theory, Signals & Systems, and Digital Logic Design.
- Guided hands-on lab sessions in microcontroller interfacing, circuit debugging, and algorithmic implementation in Python and MATLAB.

4. KEY SILICON IMPLEMENTATIONS & ENGINEERING SYSTEMS

NeuroQuant-LOB: Ultra-Low-Latency FPGA Limit Order Book & Inference Engine [*AMD Spartan-7 • Verilog • C++ • Python*]

- **Physical Silicon Verification:** Closed and deployed on physical Digilent Arty S7-25 (Spartan-7 XC7S25) on COM9 alongside an Arduino Nano hardware-in-the-loop (HIL) sidecar on COM10.
- **Timing & Silicon Footprint:** Achieved 322.06 MHz wire-speed timing closure (+0.894 ns positive slack) using **0 DSP slices** (100% fixed-point LUT arithmetic, consuming only 240 / 14,600 LUTs = 1.64% silicon footprint).
- **Ultra-Low Power:** Verified total on-chip dissipation of **82 milliwatts (0.082 W)** — yielding a 3,049x throughput/watt energy efficiency over enterprise Intel Xeon servers.
- **Deterministic SLA:** Achieved Median P50 latency = 1.230 µs and Tail P99 = 2.451 µs (exceeding institutional <10.0 µs SLA); walk-forward Deflated Sharpe Ratio DSR = 1.0000 across 500 permutations.
- **Software Suite:** Integrated automated verification harnesses, C++ pybind11 bindings, and real-time Flask/WebSocket mission control dashboard (http://localhost:5000).

PIM Edge AI Cancer Diagnostic Accelerator (DVCon India 2025) [*AMD Kintex-7 • VEGA RISC-V • Systolic Array*]

- Architected a Processing-in-Memory (PIM) AI inference accelerator integrated with India's indigenous VEGA RISC-V processor on a Genesys-2 (Kintex-7 XC7K325T) FPGA board.

- Embedded a Systolic Matrix Multiplication Engine (MME) directly into BRAM arrays to eliminate von Neumann memory bottlenecks; orchestrated DMA transfers via AXI4 memory-mapped interfaces.
- Executed quantized INT8 Convolutional Neural Network (CNN) / Vision Transformer (ViT) kernels for real-time histopathological slide classification with <20 ms inference latency and <6.8 W peak power.

USR-X: Secure, Self-Testing, Power-Gated Gear-Shift Shift Register IP [*Verilog HDL • Yosys • GTKWave • Python*]

- Designed and validated a computing register IP integrating segmented clock gating, multi-bit stride gear-shift barrel shifter, LFSR PRPG/MISR logic BIST, and stream scrambler scan chain security.
- Enabled dual-mode compute agility: operates as a bit-serial multiply-accumulate (MAC) core for MLPs/CNNs, an XNOR+popcount Binary Neural Network (BNN) accelerator, an FIR filter, and a stream cipher primitive.

5. PEER-REVIEWED PUBLICATIONS & INTELLECTUAL PROPERTY

- **Refereed International Conference Paper (Lead Author & Presenter):** Arunjyoti Sarma, Kuntal Kumar Das, Dr. L. Oppili Prasad, Dr. Ashutosh Kumar Dubey, *"Piezoelectric Nanogenerator-Based Energy Harvesting System for Low-Power Bioelectronics,"* **Proceedings of the 2026 IEEE 8th Eurasia Conference on Biomedical Engineering, Healthcare and Sustainability (IEEE-ECBIOS 2026)**, Tainan, Taiwan, October 2026. (Joint Research: Dept. of Electronics Engg. & Dept. of Ceramic Engg., IIT (BHU) Varanasi).
- **Patented Architecture & Invention Disclosure (Sole Inventor):** Arunjyoti Sarma, *"A Reconfigurable, Power-Gated Universal Shift Register with Multi-Stride Barrel Shifting and Built-In Self-Test for Bit-Serial Neural Acceleration and Cryptographic Primitives (USR-X),"* Indian & International PCT Patent Specification, Ref: USR-X/IIT-BHU/2025. (RTL verification & silicon simulation closed).
- **National Conference Technical Monograph (Lead Designer):** Arunjyoti Sarma, Swati Gupta, Updesh Kumar (Team AstraVortex, IIT (BHU) Varanasi), *"Processing-in-Memory (PIM) Hardware Accelerator Integrated with Indigenous VEGA RISC-V Processor for Real-Time Edge Cancer Diagnostics,"* **DVCon India 2025 Design Contest Technical Monograph**, Bangalore, India.
- **Competitive Research Fellowship Proposal (Principal Investigator):** Arunjyoti Sarma, *"Sub-Microsecond Deterministic Limit Order Book Engine and Neuromorphic FPGA Acceleration on Sub-Watt Silicon,"* **NVIDIA Graduate Fellowship Research Proposal & Bitter Lesson Technical Manifesto** (Certified Institute Nomination Dossier, IIT (BHU) Varanasi).
- **Journal Manuscript in Preparation (Targeted Submission):** Arunjyoti Sarma, Dr. L. Oppili Prasad, *"Multi-Physical Strain Dynamics and Electromechanical Optimization of Serpentine Interconnects on Elastomeric Substrates for Continuous Structural Health Monitoring,"* Manuscript in preparation for submission to **IEEE Sensors Journal**.

6. NATIONAL HONORS, FELLOWSHIPS & ACADEMIC DISTINCTIONS

- **Ministry of Education (MoE) Institute Doctoral Research Fellowship (2024 – Present):** Conferred full-time Institute Research Fellowship by the Ministry of Education, Government of India, to pursue advanced doctoral research in the Department of Electronics Engineering, Indian Institute of Technology (BHU) Varanasi.
- **Dual-Discipline National Engineering Mastery (GATE — EC & EE):** Four-time national qualifier in the Graduate Aptitude Test in Engineering across both Electronics & Communication Engineering (EC) and Electrical Engineering (EE) — **GATE 2024, GATE 2023, GATE 2021, and GATE 2019**; demonstrates sustained top-tier foundational competency across solid-state physics, electromagnetics, signals, communication, and digital VLSI.
- **DVCon India 2025 National Hardware Design Contest (Stage 2 Finalist Submission):** Selected to Stage 2 of the premier Design and Verification Conference (DVCon India 2025) Design Contest as Lead Architect of Team AstraVortex (IIT BHU); architected an indigenous VEGA RISC-V integrated Processing-in-Memory (PIM) accelerator on AMD Kintex-7 FPGA.
- **High-Selectivity Research Interview Honors Across Premier Institutes (2019–2024):** Shortlisted and cleared advanced technical research defense panels across India's highest-ranking doctoral institutions: Indian Institute of Science (IISc Bangalore), IIT Delhi, IIT Roorkee (MEV/RF), IIT Guwahati, IIT Kanpur, and IIT Madras.

7. TECHNICAL SKILLS & CORE COMPETENCIES

- **FPGA & Digital Hardware Design:** Verilog HDL, SystemVerilog, AMD Vivado Design Suite, Spartan-7 / Artix-7 / Kintex-7, Static Timing Analysis (STA), Clock Domain Crossing (CDC), Fixed-Point DSP-Free Arithmetic, RTL Synthesis, Yosys, Verilator, GTKWave, ModelSim, Hardware-in-the-Loop (HIL) Emulation.
- **Electronic Design Automation (EDA) & Simulation:** Cadence Innovus (P&R;, CTS), Cadence Genus (Synthesis), RTL-to-GDSII flow, COMSOL Multiphysics (Solid Mechanics, Electrostatics, Piezoelectric Devices, AC/DC), Tina-TI, LTspice, Proteus, Altium / KiCad PCB Layout.
- **Analog, Sensor & Biomedical Hardware:** Piezoelectric Nanogenerators (PVDF/PENG), Flexible Strain Sensing, Analog Front-End (AFE) Design, Instrumentation Amplifiers, Precision Bridge Readouts, MT3608 DC-DC Converters, Micro-Fabrication & Hot-Press Polymer Processing.
- **Programming & Embedded Firmware:** C, C++ (C++17/20, pybind11), Python (NumPy, SciPy, Pandas, Scikit-learn, PyTorch), MATLAB / Simulink, Embedded C, Microchip PIC, ATmega (Arduino), ARM Cortex-M, ESP32, Raspberry Pi, Linux Shell / PowerShell.
- **Architectures, Protocols & Web Telemetry:** RISC-V ISA (VEGA indigenous core), Processing-in-Memory (PIM), UART, SPI, I2C, AXI4-Lite/AXI4, FTDI High-Speed Serial, TCP/IP, WebSockets, Flask, Docker, Git, LaTeX, Signal Processing, Information Theory, Neural Networks, Real-Time Systems.